

What Do You Mean? Exploring the Alleged Theory of Mind Abilities of Large Language Models

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Abstract

This study explores the capacity of Large Language Models (LLMs) to perform tasks requiring Theory of Mind (ToM), a critical component of pragmatic language understanding. Although previous work suggests that LLMs may exhibit emergent ToM abilities, this research examines whether such capabilities genuinely involve reasoning about beliefs or merely reflect the reliance on shallow statistical cues. Through a series of controlled experiments featuring indirect speech acts and verbal irony, we assess how belief contexts influence LLM interpretations. The results reveal that, although LLMs occasionally succeed in decoding communicative intentions, their performance is not attributable to genuine ToM reasoning. This work underscores the limitations of LLMs in simulating human-like ToM and offers insight into their interpretive biases, contributing to a deeper understanding of their linguistic capabilities.¹

1 Introduction

An hallmark of human communication is the pragmatic ability to reconstruct the intended meaning of an utterance (that is, what the speaker intends to convey with it in a given situation), even when it goes beyond its literal interpretation (Grice, 1975; Levinson, 2000; Sperber and Wilson, 2002; Pinker et al., 2008). In this context, the **Theory of Mind** (ToM) – the ability to attribute mental states to others – is generally regarded to be essential for understanding intended meaning and crucial for successful social interactions in everyday life (de Villiers, 2007).

Recent advancements in Large Language Models (LLMs) and the unforeseen emergence of

new capabilities in tasks for which these models were neither explicitly trained nor instructed have prompted increased research into whether ToM could be among these newly emerging skills (Marchetti et al., 2023; Kosinski, 2024).

Determining whether these models can handle tasks that require the use of ToM would enable us to: 1) revise our understanding of ToM development from a linguistic or psycholinguistic perspective, 2) assess whether the language generated by models can be compared to human language and whether they use the same cues to perform tasks and produce texts, and 3) evaluate their effectiveness as communicators.

Kosinski (2024) claim that ToM can be regarded as one of the **emergent abilities** of large language models (LLMs). However, many of these studies overlook the conventionalization of expressions and the linguistic context, which are well-encoded in the vast amounts of text acquired during pre-training (Lu et al., 2024).

Many works in psycholinguistics (Gibbs, 1983, 2002) argue that in the presence of utterances with non-literal meanings, certain cues in the situational and linguistic context can directly lead to a non-literal interpretation, bypassing the literal interpretation stage. Thus, non-literal interpretations often become the default, particularly with conventionalized expressions. This applies to phenomena requiring mentalizing, such as **Indirect Speech Acts** (ISAs) (Gibbs, 1986b; Marocchini and Domaneschi, 2022) and **Verbal Ironies** (Ackerman, 1983; Juez, 1998; Burnett, 2015).

Building on this evidence, we hypothesize that, similar to humans, conventionalization in LLMs can play a role in enabling the completion of tasks typically considered to require mentalizing, without the need of accounting for ToM.

We aim to understand the following questions: 1) To what extent do LLMs accurately perform tasks that require mentalizing? 2) If LLMs exhibit

¹The dataset and code will be made publicly available on GitHub.

such capabilities, can their performance be shaped or enhanced by the conventionalization of linguistic expressions? 3) How do the error patterns of LLMs compare to those of humans in terms of informativeness? 4) Does the size of the language model significantly influence its accuracy in mentalizing tasks? 5) Is the performance of LLMs in these tasks indicative of their ability to simulate ToM and reasoning about varying belief states within communicative situations?

2 Related Work

Mahowald et al. (2024) argue that models employing language in human-like ways, such as LLMs, must develop two distinct competences: **formal linguistic competence** – knowledge of linguistic rules and patterns – and **functional linguistic competence** – understanding and using language in real world contexts. These competencies require the development of specialized mechanisms. This highlights the need for separate testing and evaluation of formal and functional competences through benchmarking techniques and stimuli that isolate either of them.

Our research aligns with the perspective of Hu et al. (2023), who compare language models and humans across seven pragmatic phenomena using zero-shot prompting to evaluate the models’ ability to interpret non-literal language. In contrast, our study diverges in its focus on pragmatic phenomena that require explicit **mentalizing (i.e., ToM) abilities**, that is reasoning about the others’ beliefs and intentions. Specifically, we seek to determine whether LLMs interpret non-literal language by inferring others’ beliefs, or rely on other conventionalized cues to derive the intended meaning. Existing benchmarks for assessing ToM in models have predominantly addressed social commonsense (Sap et al., 2019), situational descriptions (Nematzadeh et al., 2018; Le et al., 2019; Sap et al., 2022; Zhou et al., 2023), or social interactions (Kim et al., 2023; Li et al., 2023).

Most studies in ToM primarily investigate false belief tasks (Baron-Cohen et al., 1985), which assess whether a model can represent the beliefs of another agent that are factually incorrect but consistent with the observations of that agent (Jones et al., 2023; Shapira et al., 2023). These works focus on first or second-order ToM, while Hu et al. (2023) investigates third-order and beyond.

LLMs have been shown to succeed on some

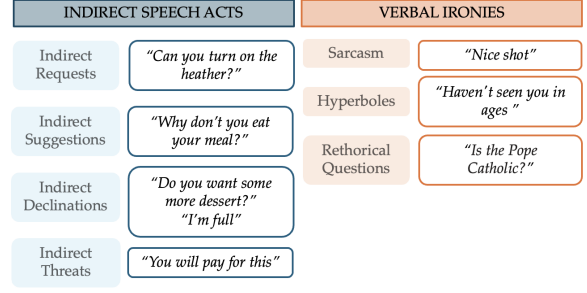


Figure 1: Linguistic phenomena tested in the present study: Indirect Speech Acts and Verbal Ironies

ToM tests (Kosinski, 2024) while failing on others (Gandhi et al., 2023; Chen et al., 2024; Ullman, 2023). As suggested in Ma et al. (2023); Chen et al. (2025); Hu et al. (2025), these varied results can be attributed both to differences in evaluation methods and to the variation in the types of prompt used in the studies, as occurs in Strachan et al. (2024). Moreover, many current findings can be explained through the use of statistical shortcuts, arising because the tasks used to investigate ToM in deep learning systems have been too narrow (Aru et al., 2023). To the best of our knowledge, the present study is the first to control the role of different belief contexts in tasks requiring ToM reasoning.

3 Experimental Materials

3.1 Linguistic phenomena

It is a common assumption that Indirect Speech Acts and Verbal Ironies require mentalizing for their highly context-dependent nature.

Indirect Speech Acts (ISAs) (Austin, 1975; Searle, 1975) occur when the speaker’s communicative intent (e.g., a request) differs from the literal meaning of their words (e.g., a question). They rely on the listener’s ability to infer the actual intent from the context and linguistic cues, often used to achieve politeness or convey subtle meanings indirectly.

We assessed four distinct types of ISAs, each differing in their level of conventionalization: **Indirect Requests** (Trott and Bergen, 2019), **Indirect Suggestions**, **Indirect Declinations**, and **Indirect Threats** (Figure 1). Accurately interpreting these ISAs may not solely depend on the ability to encode and represent others’ knowledge states and intentions, since it could also arise from con-

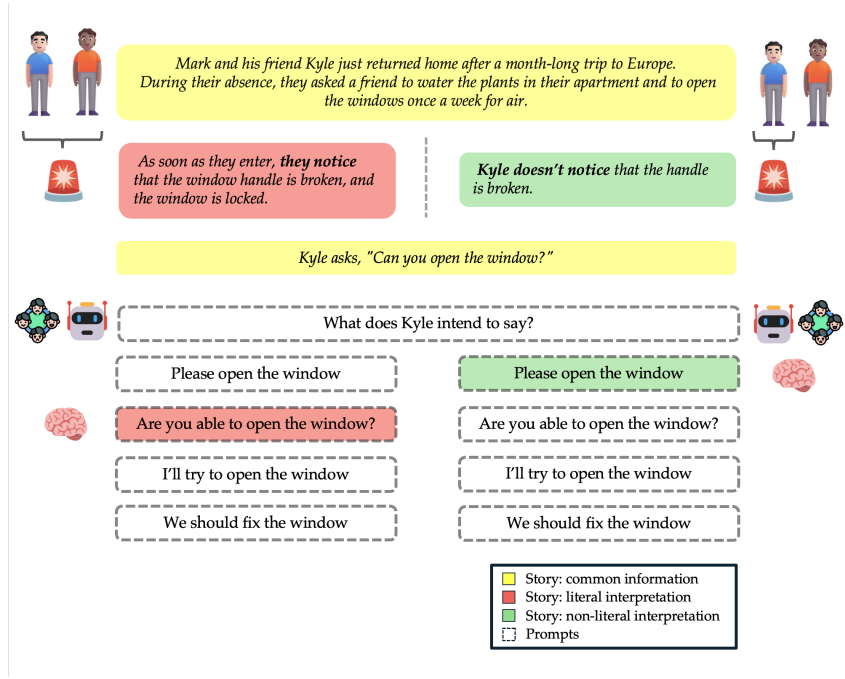


Figure 2: Stimulus from Indirect Requests. It illustrates two interpretations of the same story: non-literal (in green) and literal (in red). Each story has two versions differing for the knowledge states of the two characters. The yellow boxes highlight the common information retained across both versions. Both models and humans are presented with the same question and must select one of four possible answers.

ventional associations. In our experiments, we controlled for both literal and non-literal interpretations, so that differentiating between them really necessitates considering ToM, as **the relevant contexts differ based on the mutual beliefs held by the speaker and listener.**

Verbal irony (Skalicky, 2024) encompasses utterances where the speaker feigns ignorance of an unfulfilled expectation, whether based on personal anticipations or cultural norms. This pretense creates a discrepancy between the spoken words (reflecting the expectation) and the actual situation.

According to the graded salience hypothesis (Giora and Fein, 1999a), both literal and figurative meanings can be encoded in the lexicon based on their conventionality (e.g., *Thanks a lot* is used both sincerely and ironically).² Consequently, figurative meanings of conventional ironic expressions are more accessible and easier to comprehend than those of situation-specific ironic expressions, which may require additional contextual cues or processing. We investigated three kinds of verbal ironies: **Sarcasm** (Gibbs, 1986a), **Hyperboles** (Aguert et al., 2018; Carston and Wearing,

²Ackerman (1983) suggested that some ironic remarks might have become idiomatic, as children interpreted them ironically with minimal contextual cues.

2015), and **Rethorical Questions** (Figure 1).

3.2 Stimuli

The stimuli used in our study were inspired by the False-Belief task design, and adapted from Trott and Bergen (2019), who created pragmatically ambiguous utterances to conclude stories. In these stories, an **obstacle** to fulfil a potential request was introduced, with the speaker either present or absent when the obstacle information was disclosed. We incorporated Trott and Bergen (2019)’s stimuli on Indirect Requests into our experiment, categorizing them into two groups: IR-Os (derived from Trott and Bergen (2019) IR-Ns (created by us). Stimuli for all other phenomena were newly developed to ensure they were not included in the pretraining material.

Each story in the dataset presents two interpretations (Figure 2): in one version the belief scenario is congruent with the literal interpretation preventing any other non-literal meaning, while in other version the belief context triggers an indirect interpretation. The concluding target utterances in each story are typically conventionalized expressions of the non-literal interpretation. For Verbal Irony, conventionalization is achieved through conventional remarks following

story contexts with varying degrees of salience (Giora and Fein, 1999b): higher salience (e.g., a negative context followed by a positive outcome) or lower salience (e.g., a positive context aligned with a positive outcome).

This design enabled us to assess whether models adjusted their interpretations according to the different belief contexts of the two participants in the story. Successful task completion requires both humans and models to reason about the different epistemic states (i.e., beliefs, intentions and emotions) of the two characters. Indeed, the narrative remains consistent across both literal and non-literal versions, the only distinction being the characters’ knowledge of the obstacle. Thus, identifying the correct interpretation given a certain context requires accurate ToM reasoning. Consistent with Hu et al. (2023), we also developed reduced context versions of each story to assess whether inference relies on shallow distributional cues within the linguistic context.

The final dataset comprises 160 stories for Indirect Speech Acts, including 80 with reduced context, and 96 stories for Verbal Ironies, 46 of which have a reduced context.

4 Experimental setup

4.1 Task

We assessed both humans and models using a multiple-choice question format to enable a direct comparison between the two groups and ensure exposure to the same stimuli with minimal variations in prompts. Each story was followed by a question regarding the intended meaning of the character’s utterance, phrased as: “What does ⟨name⟩ intend to say?” Respondents selected from four options: the non-literal interpretation, the literal interpretation, and two semantically coherent but pragmatically inappropriate distractors.

4.2 Human data

We tested 360 participants using the Prolific platform. Indirect Speech Acts (ISAs) were categorized into eight groups, while Verbal Ironies were divided into four groups, with each group comprising 20-21 stories. The distribution of phenomena was randomized to ensure their equal representation and a balanced number of both indirect and direct interpretations across groups. Each group was tested with 30 speakers.

The participants were native English speakers,

Model	Parameters
GPT-2	775M
T5 + Flan-L	780M
T5-3B + Flan-XL	3B
T5-11B + Flan-XXL	11B
Tk-Instruct-3B	3B
Tk-Instruct-11B	11B
Falcon + Instruct	7B
Llama2 + Instruct	7B
Llama2-13B + Instruct	13B
Llama3 + Instruct	8B
GPT-3.5	175B (est.)
GPT-4	1.76T (est.)

Table 1: Models tested in our experiments.

equally balanced in gender, from the USA and the UK. The test included three check questions, and participants were compensated only if they answered these questions correctly (Appendix A).

4.3 Models

We evaluated a total of nineteen language models varying in architecture, size, and tuning strategy (Table 1). These include **GPT-2** (775M parameters), an autoregressive model (Radford et al., 2019); **T5-large**, **Flan-T5-large**, **T5-3B**, **Flan-T5-XL**, **T5-11B**, and **Flan-T5-XXL**, ranging from 770M to 11B parameters, with Flan variants instruction-tuned (Chung et al., 2022); **Tk-Instruct-3B** and **Tk-Instruct-11B**, instruction-tuned versions of T5 models (Wang et al., 2022); **Falcon-7B** and **Falcon-7B Instruct**, transformer-based models with 7B parameters (Almazrouei et al., 2023); **LLaMA 2-7B**, **LLaMA 2-7B Instruct**, **LLaMA 2-13B**, and **LLaMA 2-13B Instruct** (Touvron et al., 2023); **LLaMA 3-8B** and **LLaMA 3-8B Instruct** (AI@Meta, 2024); and finally, **GPT-3.5** and **GPT-4**, with estimated sizes of 175B and 1.76T parameters, respectively.³ All open-source models were accessed via Hugging Face (Wolf et al., 2020). This selection enabled us to systematically analyze how performance varies with model size, instruction tuning, and architectural family.

Each model was evaluated using the same prompt format (Appendix A), which included task instructions, the story with a question, four answer options, and the closing phrase “*The correct answer is:*”. To enhance robustness, each model received every item four times with randomized answer order. While varying the number of options does not necessarily improve accuracy, randomiz-

³Accessed via the OpenAI API (February 2024).

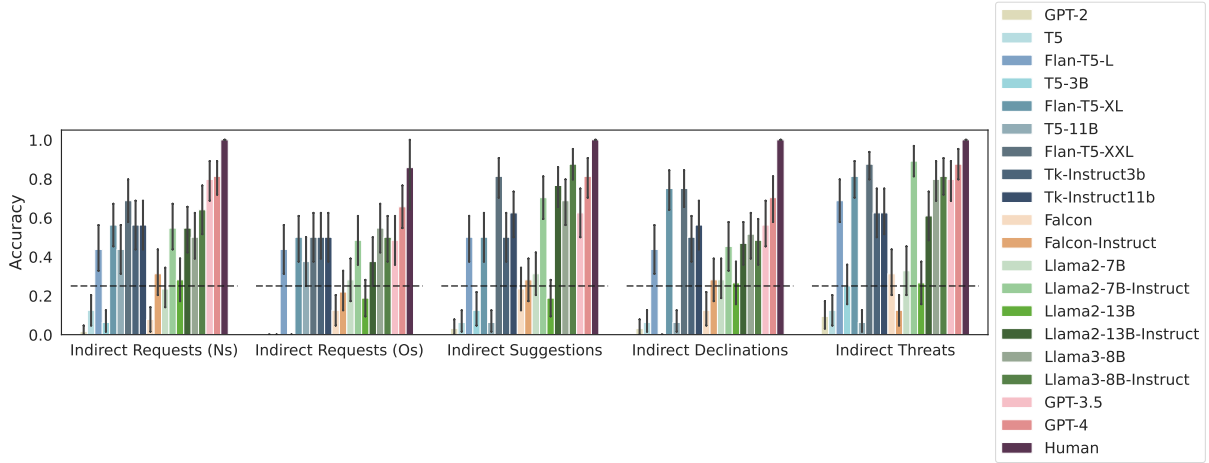


Figure 3: ISAs - Accuracy for tasks with non-reduced context.

ing and repeating responses helps strengthen evaluation reliability (Li et al., 2024).

4.4 Evaluation

We adopt a **zero-shot prompting** framework to evaluate ToM as a potentially emergent capability in language models. Recent studies identified a discrepancy between the probability models assign to word sequences, which is referred to as their “competence” since it directly relates to their language modeling training objective, and models’ “performance”, which is given by the accuracy of their responses elicited via prompting. This line of research suggests that prompting tends to underestimate the underlying linguistic knowledge of the models (Hu and Levy, 2023). Therefore, two complementary evaluation methods were used:

1. **Generation-based evaluation:** Models were prompted to produce a number (1 – 4) corresponding to the answer choice. Only valid numeric outputs were accepted, while all other outputs (e.g., text, missing numbers) were marked as invalid and considered task failures. Accuracy was computed based on exact matches to the correct answer.

2. **Probability-based evaluation:**⁴ Using the minicons library (Misra, 2022), we extracted: a) *number probability* (likelihood of the correct digit), b) *choice probability* (likelihood of the full answer sentence), and c) *continuation probability* (the probability of the answer given the prompt without the final question; see Appendix A).

⁴Probability estimates were not available for OpenAI models due to API limitations.

5 Experimental Results

5.1 Indirect Speech Acts

Figure 3 presents the accuracy scores for each type of Indirect Speech Act (ISA) across all models evaluated with the Generation method.⁵

Out of the 19 models assessed, 13 outperform the baseline accuracy of 25%, indicating that their predictions are not the result of random selection.

The best-performing models across all ISA categories include GPT-3.5 and GPT-4, Flan-T5-L, Flan-T5-XL, Flan-T5-XXL, Tk-Instruct 3B and 11B (for the T5 architecture), as well as LLaMA 2 7B Instruct, LLaMA 2 13B Instruct, LLaMA 3 8B, and LLaMA 3 8B Instruct. Notably, nearly all of these models are instruction-tuned, with the exception of LLaMA 3 8B and for each pair, the instruction-tuned version consistently outperforms its non-instruction tuned counterpart. These results suggest that instruction tuning has a greater effect on ISA performance than model size alone.

Performance varies considerably across different ISA types. For example, LLaMA 2 7B Instruct achieves the highest accuracy on Indirect Threats (0.8), but its performance drops significantly on Indirect Declinations (0.5). This pattern of variability across ISA types is observed across all models tested.

Importantly, even the best performing models fall short of human accuracy in all tasks. The ISA categories with the lowest model accuracy are Indirect Requests (Os) and Indirect Declinations, in

⁵We report generation accuracy as it is the only metric consistently available across all models, due to the lack of probability outputs for GPT-3.5 and GPT-4.

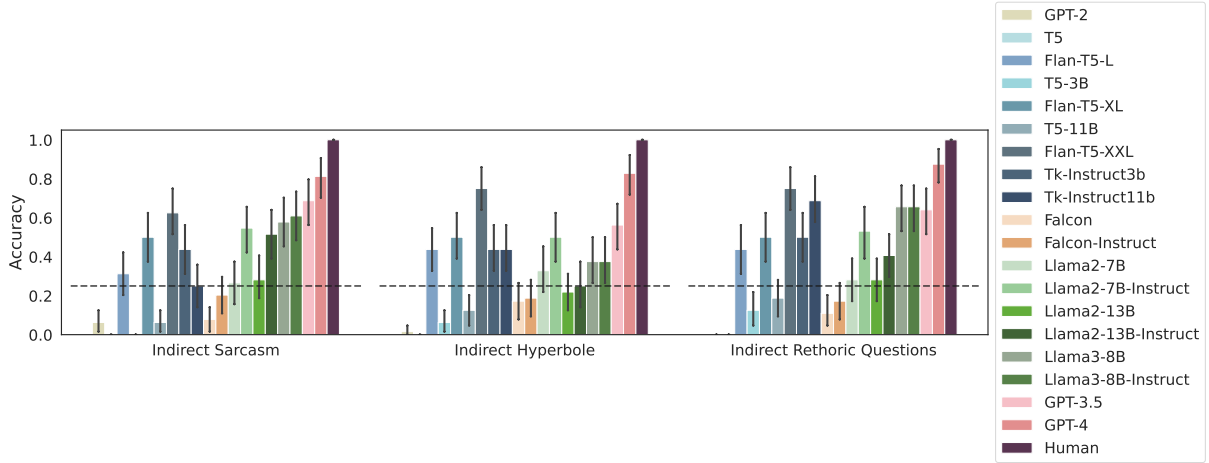


Figure 4: Irony - Accuracy for tasks with non-reduced context.

which human performance is also comparatively lower, though still consistently higher than that of the models.

Finally, tasks presented with reduced context show increased variability in human performance compared to models (see Appendix B, Figure 11).

5.2 Irony

Figure 4 indicates that LLMs exhibit similar behavior when processing irony as they do with ISAs. In particular, instruction tuning proves more impactful than model size. Nevertheless, only 9 out of the 19 models (Flan-T5-L, Flan-T5-XL, Flan-T5-XXL, Tk-Instruct 3B, LLaMA 2 7B Instruct, LLaMA 3 8B, LLaMA 3 8B Instruct, GPT-3.5, and GPT-4) outperform the baseline. Like with ISAs, model performance remains substantially below that of humans.

When context is reduced, particularly for Indirect Hyperbole and Indirect Rhetorical Questions, there is a marked decline in accuracy. However, the relative performance differences among models persist in these reduced-context settings (see Appendix B, Figure 12).

5.3 Probabilities vs. Generation

Figures 5 and 6 compare model performance across three probability-based evaluation methods. Among them, Choice Probability yields the highest performance overall, followed by Continuation Probability, while Number Probability shows minimal variation across models, suggesting it behaves as a random effect.

When comparing **Generation accuracy** with the best probability measure, that is **Choice Prob-**

ability (henceforth referred to simply as probability), we find that all models meet or exceed the baseline under the probability evaluation, which is generally better than Generation (except for Indirect Threats). This contrast may stem from smaller or non-instruction-tuned models (e.g., GPT-2, T5, T5 3B, Falcon, LLaMA 2 7B, LLaMA 2 13B, T5 11B) struggling to follow instructions during text generation. Interestingly, this pattern is also observed with Falcon Instruct, which underperforms despite instruction tuning.

Conversely, certain models that achieve the highest scores in generation, such as Flan-T5 XXL, LLaMA 3 8B Instruct, Flan-T5 XL, and LLaMA 2 7B Instruct, exhibit lower accuracy when evaluated using probability. This pattern is further illustrated in Figure 7, which shows that probability scores tend to plateau or even decline beyond a certain model size. This suggests the presence of a threshold after which generation-based evaluation outperforms probability-based measures. This discrepancy underscores the need for caution when interpreting performance metrics and highlights the importance of comparing different evaluation methods to understand their influence on results.

Notably, these trends are consistent across all ISA categories and types of irony, indicating that the choice of evaluation method systematically affects performance assessment.

6 Discussion and Analysis

In this section, we carry out a more comprehensive analysis of the experimental results to address the research questions outlined in Section 1.

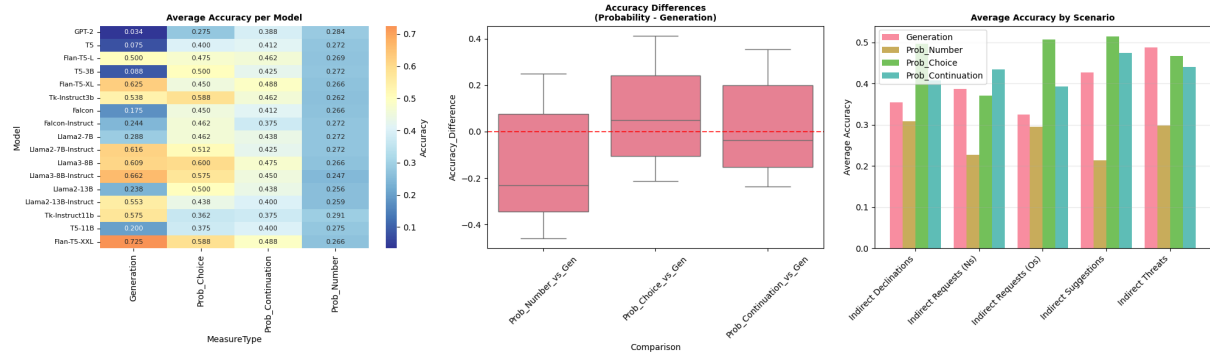


Figure 5: Comparison of all the evaluation methods in ISAs

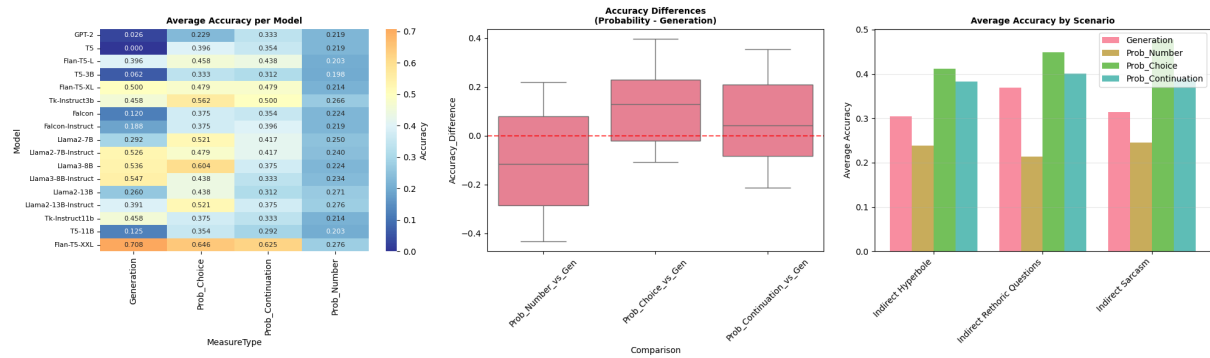


Figure 6: Comparison of all the evaluation methods in Irony

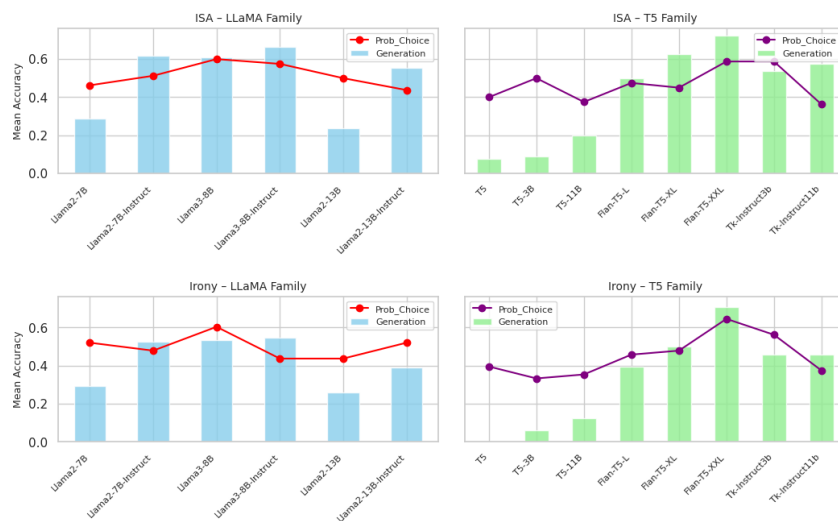


Figure 7: Comparison of Generation and Probability accuracy between the two main model families evaluated: T5-based models and LLaMA-based models.

Step	ΔR^2	F	p-value
ISA			
1 $\rightarrow$ 2	0.289	73.2	7.4e-15
2 $\rightarrow$ 3	0.001	0.29	0.59
3 $\rightarrow$ 4	0.011	2.77	0.10
Irony			
1 $\rightarrow$ 2	0.203	26.9	1.0e-6
2 $\rightarrow$ 3	0.000	0.02	0.88
3 $\rightarrow$ 4	0.011	1.49	0.23

Table 2: Model comparisons: R^2 gain and F-test statistics

6.1 LLMs and Humans in Mentalizing task

In our experiments, we controlled for both literal and non-literal interpretations. Crucially, differentiating between them necessitates considering ToM, as the relevant contexts differ based on the mutual beliefs held by the speaker and listener in the stories.

Can LLMs perform tasks that involve understanding mutual beliefs between characters? While LLMs do not achieve human-level accuracy in these tasks, certain models exhibit relatively high accuracy, particularly those that have undergone further training on specialized tasks. However, if one were to frame the question as requiring a binary answer, the response would be negative. The assessment of LLMs performance in mentalizing tasks should focus on the errors made rather than the number of correct answers. If LLMs genuinely possessed the ability to mentalize and apply such cognitive skills in task completion, their accuracy would likely approach that of humans. In humans, responses are consistently guided by their capacity for mentalizing, as this is needed to understand the knowledge contexts in the stories.

In this regard, our findings are consistent with numerous studies in the scientific literature (Ullman, 2023), which underscore the limitations of LLMs in executing tasks that necessitate mentalizing. However, if true ToM-like capabilities are still lacking in LLMs, what mechanisms or factors account for the particularly accurate task completion observed in several cases?

6.1.1 Statistical Analyses of the Models' Accuracy

To answer this question, we conducted a series of additional statistical analyses to explore the respective contributions of size, instruction tuning,

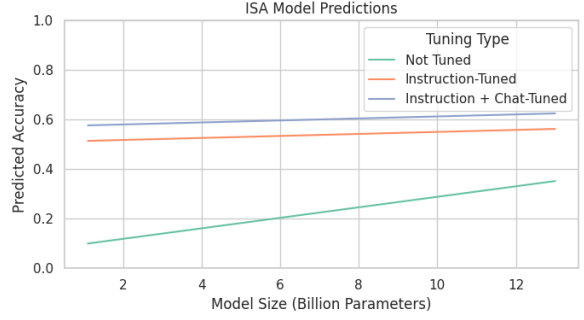


Figure 8: Results of a four-step regression analysis examining the impact of model size, instruction tuning, and chat tuning on accuracy for ISA.

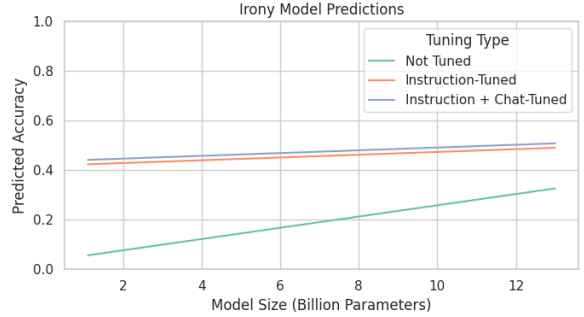


Figure 9: Results of a four-step regression analysis examining the impact of model size, instruction tuning, and chat tuning on accuracy for Irony.

and chat tuning to model accuracy.⁶ As shown in Figure 7, instruction-tuned models consistently outperform their base versions across both the T5 and LLaMA model families, suggesting that the ability to follow instructions plays a critical role in task performance.

To better quantify these effects, we conducted a four-step regression analysis with increasing factor complexity. In the first step, we examined whether model size alone could account for variation in accuracy, measured with the **Choice Probability** method. The second step introduced a binary variable for instruction tuning, allowing us to measure its additional impact beyond size. In the third step, we added a chat tuning variable to evaluate whether conversational fine-tuning provided further improvements. Finally, in the fourth step, we included an interaction term between size and instruction tuning to explore whether the effect of model size depends on the presence of instruction

⁶We excluded GPT-3.5 and GPT-4 from this analysis due to the unavailability of intermediate-sized variants and limited computational resources, which constrained our ability to include these models in the regression study.

tuning. This stepwise approach allowed us to isolate the relative influence of each factor and examine how they interact in shaping model accuracy.

Figures 8 and 9 and Table 2 provide further support for the central role of instruction tuning in determining model performance. Compared to model size, instruction tuning shows a significantly stronger effect on accuracy across both ISA and irony tasks. Additionally, chat tuning and interaction effects do not provide statistically meaningful gains in either task (see F-tests), suggesting that the main performance gains come from the initial instruction tuning step rather than further refinement through chat-specific tuning.

Such results reinforce the conclusion that instruction tuning is a key factor influencing model effectiveness, surpassing the impact of scale alone. This also calls for caution in treating the alleged pragmatic abilities of LMs as emergent properties of their predictive training objective (Kosinski, 2024). Rather, a key factor in shaping such abilities lies in the highly sophisticated supervised post-training processes that instruction-tuned models undergo.

6.1.2 Investigating the Role of Fine-tuning

As noted in Section 5, there are clear differences in performance across phenomena for each model. For example, LLaMA 2 7B Instruct achieves notably higher accuracy on Indirect Threats than on Indirect Declinations, and even outperforms the larger LLaMA 2 13B Instruct on the same task. A similar pattern emerges for Indirect Hyperboles in the irony tasks. Overall, models tend to perform better on ISAs than on irony, with irony consistently showing lower accuracy scores.

Interestingly, across all sizes of the Flan Instruct models, accuracy for Indirect Threats is higher than for other ISA categories. Among them, Flan-T5 XXL stands out as the top-performing model on irony tasks. Another notable exception is Tk-Instruct 11B, which shows higher accuracy on Indirect Rhetorical Questions compared to other types of irony.

Despite these variations, the general performance trends between the T5-based and LLaMA-based model families remain largely aligned across phenomena. These patterns raise the question of whether such variations may be explained by differences in the fine-tuning datasets used during instruction or chat tuning.

The Flan model family was fine-tuned on 473

diverse datasets across 146 task categories, mainly focused on standard NLP tasks like question answering, dialogue, and reasoning, with no explicit emphasis on ISAs or irony. This suggests that model performance in these areas might stem from pretraining rather than fine-tuning.

T5 models are pretrained on the Colossal Clean Crawled Corpus (C4), a large collection of web pages with natural language. To evaluate its coverage of ISAs and irony, we conducted a similarity search for conventionalized expressions within a 1-million-document sample, using a 90% similarity threshold across sliding windows.

Our preliminary analysis shows that Indirect Threats have a 0.01% match rate within the pretraining corpus, whereas other phenomena register no matches at all.⁷ Despite its simplicity, this analysis helps explain the slight performance advantage observed for Indirect Threats, suggesting they are relatively more **conventionalized**, that is, **they are likely to be more frequently encountered during pretraining**, giving models increased exposure to them compared to other ISAs. Crucially, these expressions appear in the same form as those used in our tasks, reinforcing their classification as conventionalized patterns.

Given that Tk-Instruct models are also based on pretrained T5 architectures, it follows that differences in fine-tuning data or procedures must play a significant role in explaining the performance gap observed between Flan and Tk models. Tk-Instruct is trained using the tasks and instructions in Natural Instructions benchmark, which contains 1600+ tasks in 70+ broad categories in total, we assume that these different fine-tuning materials is the reason of the difference with Flan series. What we struggle to explain is the difference between the 3B and 11B version in Indirect Sarcasm where the 3B version performs better than 11B.

LLaMA 2 models (7B and 13B) were trained using publicly available instruction datasets, followed by reinforcement learning with human feedback (RLHF). While Touvron et al. (2023) do not specify the exact instruction datasets used, it is reasonable to assume that both versions were fine-tuned on the same data. This makes it difficult to account for the consistent performance advantage of the 7B model over the 13B version on tasks

⁷This result is based on a basic regular-expression-based similarity check; more advanced techniques would likely uncover more detailed patterns.

such as Indirect Threats and Indirect Hyperboles. In contrast, LLaMA 3 8B Instruct incorporates several updates over its predecessor, including the use of Direct Preference Optimization (DPO) rather than Proximal Policy Optimization (PPO). However, this architectural refinement does not yield a dramatic performance improvement over LLaMA 2 Instruct models or even the base LLaMA 3 8B, suggesting that observed gains may be more attributable to enhancements in pretraining rather than fine-tuning alone. Since Meta has not released the pretraining data, it remains challenging to determine the exact source of these differences.

Falcon models are fine-tuned on a relatively modest 250-million-token mix of chat and instruction-oriented data. The diversity of task types is also narrower than in other model families, drawing from sources such as Baize (chat-focused), GPT4All (instructional), GPTeacher (instructional), and RefinedWeb-English (web-based filler content). This more limited fine-tuning set may contribute to the model’s relatively weaker performance.

It is important to emphasize that fine-tuning does not introduce entirely new information but rather enhances the model’s ability to utilize information already acquired during pretraining. The observed performance differences between base and instructed models thus reflect not the presence or absence of information, but the model’s capacity to interpret and apply it effectively. This distinction also helps explain discrepancies between generation-based and probability-based evaluations: The relevant patterns may be present in the base models, but the ability to recognize and follow task-specific instructions is closely tied to instruction tuning.

6.2 Do Belief Contexts Really Drive LLMs’ Interpretation?

The key design feature of our test items is the difference in the belief contexts of the two characters triggering either the literal or the non-literal interpretation of the target sentences. We refer to these as **belief-biased contexts**. A key question, then, is: **Does a genuine causal relationship exist between such biased contextual belief states and the answers produced by LLMs?**

Our findings suggest that reducing contextual information does not result in different interpretations by the model (full context vs. short con-

ANOVA (Label)	$F(2, 17405) = 11.635, p < 0.00001$
Tukey HSD Post-hoc	
ToM vs. ToM-N	$p = 0.454, \text{n.s.}$
ToM vs. ToM-0	$p < 0.001, \text{significant}$
ToM-0 vs. ToM-N	$p = 0.005, \text{significant}$

Table 3: One-way ANOVA and Tukey HSD post-hoc comparisons

text, Appendix B). Therefore, to explore this issue we conducted additional experiments designed to investigate whether the presence or absence of belief-related context in stories influences a model’s ability to select the correct interpretation.

To this end, we introduced two new test sets: i.) **ToM-N**, in which the target utterances are preceded by a *neutral context without belief states likely to trigger either a literal or non-literal reading* (see Appendix B), and ii.) **ToM-0**, in which the target utterances are *without any contextual information*. The original stimuli with belief-biased context requiring ToM reasoning are henceforth designated as **ToM**.

A one-way ANOVA conducted on the models’ Choice Probability accuracy reveals a significant effect across the three context (belief-biased, belief-neutral, and zero) conditions ($F(2, 17405) = 11.64, p < .001$). Post-hoc Tukey HSD comparisons (Table 3) indicated that the ToM-0 condition differed significantly from both ToM-N and ToM, whereas the difference between ToM and ToM-N was not significant. This suggests that models are **insensitive to the belief states variation in the context**, since the model behavior does not substantially change in the belief-neutral and in the belief-bias conditions. By contrast, the significant difference between ToM-N and ToM-0 indicates that providing situational context leads to a shift in interpretation, decreasing the likelihood of a literal reading (as reflected in the mean values in Table 5).

We conclude that the non-literal interpretation of target sentences is driven by statistical features in the context, but it cannot be causally attributed to a genuine ToM-like sensitivity to the characters’ differing belief states.

6.2.1 The Literal Bias of LLMs

Models show a consistent bias toward literal interpretations. In fact, the accuracy with the ToM stimuli is higher in literal contexts (Table 4). Interestingly, stimuli were designed such that, the non-literal interpretation of target sentences is the

Model	ISAs		Irony	
	Literal	Non-Literal	Literal	Non-Literal
Falcon	0.181	0.169	0.104	0.135
Falcon-Instruct	0.413	0.075	0.260	0.115
Flan-T5-L	0.675	0.325	0.708	0.083
Flan-T5-XL	0.450	0.800	0.750	0.250
Flan-T5-XXL	0.725	0.725	0.917	0.500
GPT-2	0.019	0.050	0.031	0.021
GPT-3.5	0.619	0.688	0.792	0.469
GPT-4	0.756	0.788	0.875	0.802
Human	1.000	0.947	1.000	1.000
Llama2-13B	0.275	0.200	0.344	0.177
Llama2-13B-Instruct	0.575	0.531	0.615	0.167
Llama2-7B	0.344	0.231	0.438	0.146
Llama2-7B-Instruct	0.900	0.331	0.885	0.167
Llama3-8B	0.744	0.475	0.719	0.354
Llama3-8B-Instruct	0.800	0.525	0.812	0.281
T5	0.000	0.150	0.000	0.000
T5-11B	0.200	0.200	0.083	0.167
T5-3B	0.175	0.000	0.083	0.042
Tk-Instruct11b	0.575	0.575	0.708	0.208
Tk-Instruct3b	0.875	0.200	0.792	0.125

Table 4: Accuracy by interpretation type (Literal vs. Non-Literal) across models for ISAs and Irony in ToM context.

most typical and strongly conventionalized one. Instead, the pattern suggests that models default to literal interpretations, as reflected in their lower accuracy for non-literal contexts.

To examine variation in interpretations across context types, we employed a soft clustering method based on Expectation Maximization (EM) with a Gaussian Mixture Model. The analysis was constrained to two clusters, labeled **Literal** and **Non-Literal**, which were determined from the distribution of model-assigned Choice Probabilities for the non-literal answer option in the ToM literal and ToM non-literal conditions (corresponding to the two possible interpretations). The EM algorithm estimates these distributions and assigns posterior probabilities to new data points, allowing us to assess how interpretations from ToM-N (neutral context) and ToM-0 (no context) align with either the Literal or Non-Literal cluster. This approach provides a probabilistic estimate of whether a given target sentence in the belief-neutral or zero-context conditions is interpreted more like the non-literal or literal cases in the belief-biased ToM condition.

Figure 10 shows that both ToM-0 and ToM-N contexts align more closely with the Literal cluster, indicating a model preference for literal interpretations in the absence of explicit belief-biased contextual cues. Further insight comes from the Δ **Mean**, defined as

$$\Delta = \alpha - \beta \quad (1)$$

where α denotes the model’s probability of a non-

Condition	Mean Δ
ToM	−0.0113
ToM-N	−0.0131
ToM-0	−0.0186

Table 5: Mean Δ accuracy across conditions

literal interpretation and β the probability of a literal interpretation. This metric quantifies the relative preference for non-literal versus literal readings. Positive values indicate a bias toward non-literal interpretations, while negative values indicate a bias toward literal interpretations.

As shown in Table 5, all mean Δ values are slightly negative, indicating a general preference for literal interpretations. The **ToM-0** condition exhibits the strongest literal bias in the absence of context, which diminishes in **ToM-N** and **ToM**, with the latter showing the weakest bias. The significant difference between ToM-N and ToM-0 (Table 3) suggests that even a belief-neutral context can reduce this bias.

This generalized **literal bias** of LLMs across different context types is rather surprising given our stimuli design, which was structured with highly conventionalized non-literal readings and therefore expected to be selected as the default ones. Importantly, the LLM behavior also contrasts with human one. According to the **conventionalization hypothesis** (Gibbs, 1983, 2002) humans preferentially adopt the most conventionalized interpretation of an expression, typically the non-literal one in established ISAs. If non-literal readings constitute a statistical default, models should tend to interpret utterances such as “*Can I open the window?*” as indirect requests. In contrast, models consistently favor literal readings, highlighting a fundamental difference from human language processing.

7 Conclusions

Previous studies have reported impressive performances of LLMs on psycholinguistic tasks designed to require ToM. This paper critically assesses these claims by examining whether LLMs can infer the intended meaning in two pragmatic tasks: ISAs and irony. We used short narratives in which the correct interpretation, literal or non-literal, depended on reasoning about the differing mental states of two characters, allowing us to

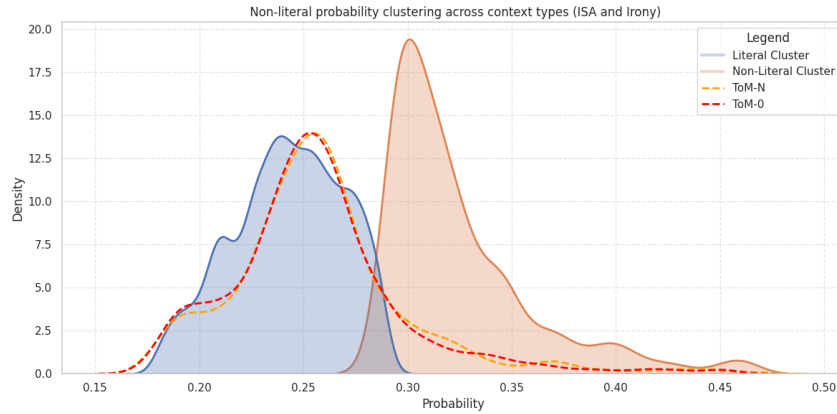


Figure 10: Non literal probability clustering across context types for ISA and Irony

test whether models really adjust their interpretations based on belief contexts. We evaluated nineteen models, including instruction-tuned variants, across four contextual conditions.

Our results reveal several key findings:

1. LLMs consistently underperform compared to humans in the pragmatic tasks tested in the experiments.
2. LLMs struggle to correctly interpret non-literal meanings, even when such interpretations are more conventionalized than their literal counterparts, showing a clear literal bias.
3. Model interpretation of target utterances appears largely unaffected by the belief context, suggesting that even correct responses may not reflect genuine ToM abilities.
4. Generic contextual information (e.g., descriptive or neutral situational cues) influences interpretation more than belief contexts, suggesting that the model behavior is driven by surface statistical clues in the context, rather than by the actual belief states of the characters in the story.

Our analyses further reveal that while instruction tuning and increased model size improve performance on tasks requiring reasoning about characters and beliefs, these improvements alone are not sufficient to simulate ToM-like capabilities. Specifically, models display a persistent difficulty in identifying non-literal meanings, frequently applying literal interpretations regardless of the contextual requirements.

Overall, our findings underscore a critical distinction between human and model behavior in

interpreting intended meanings. Unlike humans, who often default to non-literal interpretations in stereotypical contexts, models tend to favor literal interpretations across all conditions. This suggests that LLMs may rely on shallow pattern-matching rather than true inferential reasoning. As such, attributing ToM abilities to model outputs without considering alternative explanations is problematic. Extreme caution is therefore warranted when interpreting the behavior of LLMs as evidence of ToM-like reasoning. The observed performance may instead reflect systematic biases in the models’ interpretive strategies, shaped by pre-training and fine-tuning data distributions rather than belief-based reasoning.

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A Appendix - Stimuli and Prompts

In Table 6 we present stimuli for each phenomena in our dataset. Stimuli were presented both to model and humans with slight changes. We tested humans through the Prolific platform with the following premise:

In this test, you will read short stories and you will have to answer a question about one character of the story. READ EACH STORY CAREFULLY! You must chose one among 4 possible answers. Choose the answer that you think to be correct given the story. The test contains thee check questions. IMPORTANT. The participant will not be paid unless it answers to these questions correctly.

The check questions included in stimuli submitted to humans were the following:

Today is Sunday and after school Tommaso goes out to play with Red in the garden. When he hears Lisa call "Lunch is ready, time to stop playing now and come in". He runs in and sits at the table.

What does Lisa inted to say?

1 The lunch is ready; 2 The lunch isn't ready; 3 I want to go out fot lunch; 4 It's time for playing.

"Amy!" Mrs. Grove called from the door that opened towards the garden. But no answer came.

What does Mrs. Grove intend to say?

1 She is asking a question; 2 She is calling Amy; 3 She is answering a question; 4 She wants to change her name.

My boyfriend David, is a dentist. David says that he is happy to work in the centre of a small city. He can go shopping in his lunch hour, or go for a walk and look at the ducks on the river. He doesn't need a car, unless he wants to drive out of York into the countryside.

What does David intens to say?

1 He is happy to work in the centre of a small city; 2 He doesn't like working in a small city; 3 He wants to move abroa; 4 He hates the small cities.

The sequence of stories within the form, as well as the options presented in each story, were randomized. This randomization minimizes subjective error in human psycholinguistic text construc-

tion, as the entire process of form creation and stimuli randomization was automated. To generate forms from the table, we employed App Scripts. The Excel files containing human responses, along with the order of stimuli and samples, are stored on GitHub.

The query used to prompt the models, instead, was the following (example extraxted from indirect declinations):

"Task: I'll give you a story and I'll ask you to answer a question about one character of the story. I'll give you four possible answers to the question and you must choose the right one. The possible answers are 1, 2, 3 and 4. Story: I invited my brother Kevin to dinner to celebrate my birthday. Kevin and I rarely get to spend time together. For the occasion I prepared a fish dinner, and I bought some excellent Italian wine, our favorite. After a couple of drinks, I ask Kevin to pass me the glass so I can pour him more wine, but he tells me: "I need to drive". What does Kevin intend to say? Options: 1 Please, do not pour me more wine. 2 I drive tonight 3 I don't like this wine. 4 We should drink French wine. The correct answer is:"

The query used to collect continuation probabilities is:

"Task: I'll give you a story and I'll ask you to answer a question about one character of the story. I'll give you four possible answers to the question and you must choose the right one. The possible answers are 1, 2, 3 and 4. Story: I invited my brother Kevin to dinner to celebrate my birthday. Kevin and I rarely get to spend time together. For the occasion I prepared a fish dinner, and I bought some excellent Italian wine, our favorite. After a couple of drinks, I ask Kevin why he stopped drinking. Kevin replies "I need to drive". What I understand is Options: 1 that Kevin is asking me to do not pour him more wine. 2 that Kevin has to drive tonight 3 that Kevin doesn't like this wine. 4 that we should drink French wine . The correct answer is:"

B Appendix - Reduced Contexts

Figures 11 and 12 show generation accuracy for tasks with reduced contexts.

An example of ToM-N is:

Mark and his friend Kyle just returned home after a month-long trip to Europe. During their

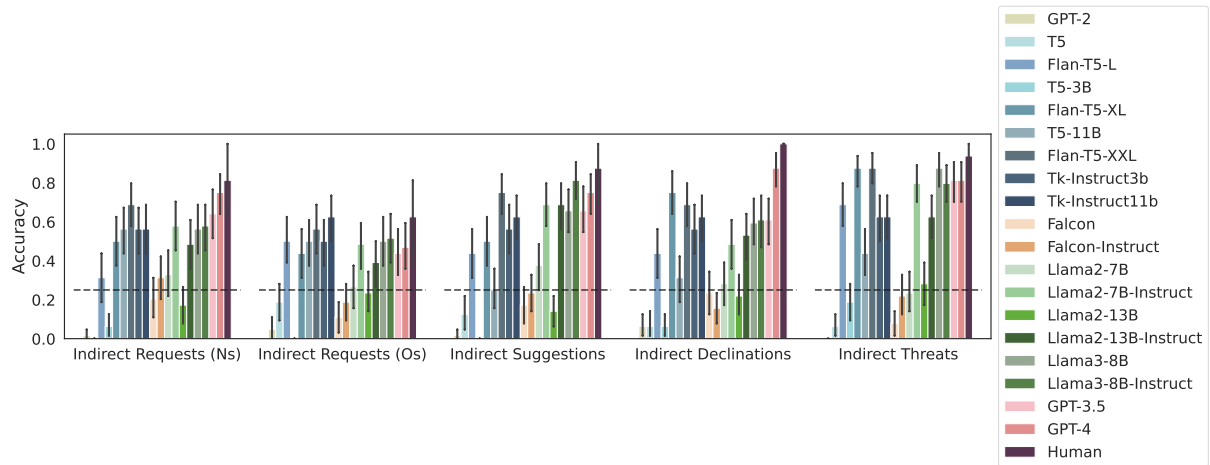


Figure 11: ISAs - Accuracy for tasks with reduced contexts.

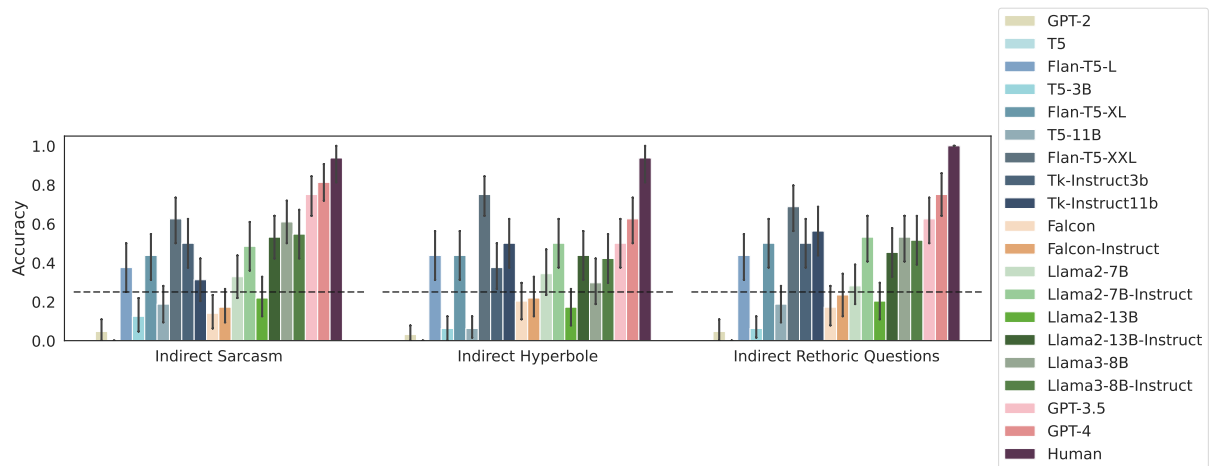


Figure 12: Irony - Accuracy for tasks with reduced contexts.

absence, they asked a friend to water the plants in their apartment and to open the windows once a week for air. As soon as they enter, Kyle asks, "Can you open the window?".

An example of ToM-0, instead, is:

Kyle asks, "Can you open the window?".

Phenomenon	Literal Story	Non-Literal Story	Question	Options
Indirect Request (Os)	You and your friend Jonathan are taking a road trip. You began in California, and are now passing through Michigan. It's almost winter, so it's very cold outside-especially for Southern California dwellers like you and Jonathan. You see that you're almost out of gas, so you stop at a gas station in a small town. You fill up the tank, and then the two of you go inside the gas station to buy some water and snacks. When you return to the car and start up the engine, you and Jonathan notice with some dismay a blinking light, which indicates that the car's heating system is broken. You both bundle up. As you leave the station, Jonathan shivers in his seat. He turns to you and says, "Man, it's really cold in here"	You and your friend Jonathan are taking a road trip. You began in California, and are now passing through Michigan. It's almost winter, so it's very cold outside-especially for Southern California dwellers like you and Jonathan. You see that you're almost out of gas, so you stop at a gas station in a small town. While you fill up the tank, Jonathan goes inside to buy some water and snacks. As you're checking the meter, you notice with some dismay a blinking light, which indicates that the car's heating system is broken. You finish filling up the gas and wait for Jonathan. Jonathan returns with some snacks, and you both set off. As you leave the station, he shivers in his seat. He turns to you and says, "Man, it's really cold in here."	What does Jonathan intend to say?	1. Could you turn on the heater? 2. I'm really cold; it's too bad the heater is broken. 3. We should fix the heater now. 4. We should get out of the car to warm up.
Indirect Suggestions	Noah lives in New Jersey, but his girlfriend Rachel lives in California. Noah often complains to his friend Jordan that he never gets to spend time with Rachel and that he misses her a lot. Today, Jordan and Noah are out to lunch and Noah is talking about how he decided to quit his job and move to Florida. Jordan is surprised by this decision and says: "Why don't you move to California?"	Noah lives in New Jersey, but his girlfriend Rachel lives in California. Noah often complains to his friend Jordan that he never gets to spend time with Rachel and that he misses her a lot. Today, Jordan and Noah are out to lunch and Noah is complaining about his long-distance relationship. Jordan knows that Noah would really like to move in with Rachel and that he isn't fully satisfied with his life in New Jersey. Jordan at one point says: "Why don't you move to California?"	What does Jordan intend to say?	1. I suggest you move to California. 2. Why are you moving to Florida and not California? 3. You shouldn't move to California. 4. California is uglier than Florida.
Indirect Declinations	I invited my brother Kevin to dinner to celebrate my birthday. Kevin and I rarely get to spend time together. For the occasion I prepared a fish dinner, and I bought some excellent Italian wine, our favorite. After a couple of drinks, I ask Kevin why he stopped drinking. Kevin replies "I need to drive".	I invited my brother Kevin to dinner to celebrate my birthday. Kevin and I rarely get to spend time together. For the occasion I prepared a fish dinner, and I bought some excellent Italian wine, our favorite. After a couple of drinks, I ask Kevin to pass me the glass so I can pour him more wine, but he tells me: "I need to drive".	What does Kevin intend to say?	1. Please, do not pour me more wine. 2. I drive tonight. 3. I don't like this wine. 4. We should drink French wine.
Indirect Threats	Janet and Monica are planning to go on a trip to the lake. They decide to invite their friend Brittany too. However, Brittany decides to join them after the picnic at Janet's house. Janet asks if she wants them to pick her up before they go home, but Brittany replies: "I know where you live".	Janet and Monica are planning to go on a trip to the lake. They decide to invite their friend Brittany too. However, Brittany decides to join them after the picnic at Janet's house. Once at Janet's house, Brittany reminds Janet that she has to pay her back the money she lent her and adds "I know where you live."	What does Brittany intend to say?	1. Give me my money back or I will come looking for you. 2. I know where your house is. 3. I really like your house. 4. I don't want to come to your house.
Sarcasm	Kyle and Mark play soccer on a team in their town. Today their team is playing the final. Kyle and Mark both play offense. During the match Mark makes a pass to Kyle and Kyle scores a goal. Mark tells him "Nice shot."	Kyle and Mark play soccer on a team in their town. Today their team is playing the final. Kyle and Mark both play offense. During the match Mark makes a pass to Kyle but Kyle misses the shot and fails to score a goal. Mark tells him "Nice shot."	What does Mark intend to say?	1. It was a really bad shot. 2. It was a really good shot. 3. You should try again. 4. You should take a shot.
Indirect Hyperbole	Monica and Rachel are two old friends who haven't met since university. After speaking by phone, they decide to meet for a coffee to see each other again and update on their lives. When they meet, Monica says "Haven't seen you in ages."	Monica and Rachel are two best friends. They went to the same high school and the same university, live in the same house, and practically spend all their time together. Tonight, Monica decides to go out with some of her friends and while she is at a restaurant she meets Rachel who is having dinner with her boyfriend. Monica tells Rachel "Haven't seen you in ages."	What does Monica intend to say?	1. We always see each other. 2. Haven't seen each other for a long time. 3. I don't want to see you again. 4. I would like to see you again soon.
Rhetorical Questions	Clara attends the art academy and often during the courses they must copy famous paintings. During today's lesson, Clara must reproduce a painting of her teacher. Clara is sitting in the last row and cannot distinguish the colors of the painting well, also because the colors are very dark. After drawing the draft, she begins to color, but is undecided about which are the right tempera colors to use, so she says to the teacher "Is the sky blue?"	Clara attends the art academy and often during the courses they must copy famous paintings. During today's lesson, Clara must reproduce a painting of her teacher. Clara is sitting in the last row and cannot distinguish the colors of the painting well, also because the colors are very dark. After drawing the draft, she begins to color. At a certain point one of her classmates asks Clara if she will use the brush to color and Clara says, "Is the sky blue?"	What does Clara intend to say?	1. The answer is obvious. 2. I want to know if the color of the sky is blue. 3. I think the sky is blue. 4. I don't think the sky is blue.

Table 6: Stimuli for each phenomenon, illustrating the two versions of the story. Literal correct answers are with number 2 and non-literal correct answers with number 1.